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Vellore Institute of Technology
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School of Computer Science Engineering and Information Systems

Department of Software and Systems Engineering

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ISWE209L – Data Mining (Course Project)

Review – 01

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Faculty - Dr. Ramkumar

Team Members :

Dhanush M (23MIS0526)
Dhanush T (23MIS0404)
Kishore E (23MIS0494)
Sakthi Maran A (23MIS0298)

Introduction

Mental health is a key factor in overall well-being, yet issues like stress, anxiety, and depression are increasingly common. Early detection of mental health risks can help in providing timely support and interventions. This project predicts “**mental health risk levels using lifestyle, psychological, and social factors such as sleep, activity, depression, and anxiety scores**”. Through feature engineering, clustering, PCA, and visualization, the model identifies patterns that improve prediction accuracy. The goal is to create a reliable, data-driven system that supports students, employees, and healthcare providers in promoting better mental health outcomes.

Abstract (min 200 words)

Mental health risk prediction is essential for early detection and timely intervention. This project uses a dataset containing demographic, lifestyle, psychological, and social features such as age, sleep hours, physical activity, depression and anxiety scores, productivity, and social support. Data preprocessing, outlier removal, and feature construction were applied to enhance data quality. PCA-based dimensionality reduction and clustering were performed to identify patterns among individuals. Predictive models classify individuals into low, medium, and high-risk categories. Results show that combining psychological scores with lifestyle indicators improves accuracy, supporting mental health awareness and preventive strategies through data-driven insights.

Dataset Description & Source

The project utilizes a dataset titled "mental_health_data.csv" for the analysis and prediction of mental health risks. This dataset was collected from survey responses and self-reported measures, providing insights into lifestyle, psychological, and social factors influencing mental well-being. The dataset includes the following key features, each providing a different facet of information:

The dataset includes the following key features:

- **Demographics:** age, gender, and employment_status provide basic personal information relevant to risk segmentation.
- **Mental Health History:** mental_health_history and seeks_treatment indicate whether individuals have prior issues or are actively seeking help.
- **Lifestyle Factors:** sleep_hours and physical_activity_days highlight daily habits that significantly influence mental health.
- **Psychological Metrics:** depression_score, anxiety_score, and productivity_score capture key indicators of stress, emotional health, and functional performance.
- **Social Factors:** social_support_score represents the level of support individuals receive from friends, family, or peers.
- **Target Variable:** mental_health_risk classifies individuals into categories such as Low, Medium, or High risk based on their responses.

This curated dataset is highly suitable for a data mining project focused on predictive modeling of mental health risks, as it contains a balanced mix of categorical and numerical features. These variables allow for building models that can detect risk levels and provide actionable insights for early intervention.

Feature Engineering – Work flow carried out

1. Data Cleaning
2. Understanding of relationship among features
3. Hypothesis testing
4. Feature Selection
5. Dimensionality reduction
6. Feature construction
7. Implementing various proximity measures
8. Visualization
9. Curated dataset and inference

Github link:

[https://colab.research.google.com/drive/1UD06keb2ADuImc-iBdicQgeygW3MwTiX10.](https://colab.research.google.com/drive/1UD06keb2ADuImc-iBdicQgeygW3MwTiX10)
[Target Data mining task focused & Justification](#)

Justification for Target Task

Business / Social Objective Alignment

- The goal of this project is to predict mental health risks among individuals using lifestyle, behavioral, and psychological factors.
- Early detection of mental health risks enables timely intervention, better allocation of healthcare resources, and improvement in overall productivity and well-being.
- Institutions (universities, companies, and healthcare providers) can use these predictions to design preventive strategies (counseling, awareness programs, stress management workshops).

Demand Analysis

- Features like sleep_hours, physical_activity_days, stress_level, depression_score, and anxiety_score are directly linked to mental health conditions.
- By analyzing these features, the project can detect high-risk patterns (e.g., poor sleep + high anxiety = higher risk).

Customer / Patient Segmentation

- Features such as age, gender, employment_status, and work_environment allow segmentation of individuals into meaningful groups.
- Different intervention strategies can be designed for different segments:
- Students → focus on academic stress and counseling.
- Employees → focus on workplace stress and work-life balance.
- Unemployed → focus on financial and social support.

Dynamic Adjustments

- Constructed features like mental_health_index and productivity_per_symptom help in quantifying the severity of conditions.
- This allows the model to dynamically update the risk level of individuals based on their latest inputs (e.g., drop in productivity + rising anxiety).

Time-Based Insights

- The LastUpdatedTimestamp (or periodic survey timestamps) enable temporal tracking of mental health trends.
- This helps identify:
- Periods of increased stress (e.g., exams, deadlines, festive seasons).
- Seasonal variations in mental health risk (e.g., higher depression rates in winter).

External Factor Impact

- Features like work_environment (remote, hybrid, onsite) and social_support_score capture external influences.
- These ensure the model accounts for environmental and social conditions that may either worsen or improve mental health.

Clustering for Similarity Analysis

- Clustering groups individuals with similar risk patterns (e.g., high stress + low sleep vs. moderate stress + good sleep).
- This enables targeted interventions for each cluster instead of treating all individuals in the same way.

Revenue / Impact Maximization

- Though the aim is social rather than financial, accurate predictions lead to:
- Reduced healthcare costs (early intervention avoids severe cases).
- Higher productivity in workplaces and educational institutions.
- Improved overall quality of life and reduced absenteeism.

How the Project Supports the Target Task

1. Data Cleaning

- Handles missing values, duplicates, and outliers in sensitive health data.

2. Feature Engineering

- Constructs meaningful features like `mental_health_index` and `productivity_per_symptom` for better prediction.

3. Proximity Measures & Clustering

- Identifies similarity patterns between individuals, simplifying intervention strategies.

4. Visualization & Insights

- Heatmaps, scatter plots, and cluster plots reveal hidden patterns (e.g., link between sleep and anxiety).

5. Hypothesis Testing

- Validates the significance of factors like `employment_status` or `social_support_score` on risk levels.

6. Dimensionality Reduction (PCA)

- Reduces complexity while retaining essential information for efficient model training.

7. Curated Dataset

- Provides a clean dataset with all relevant features for mental health risk prediction.

Conclusion

The target data mining task of Mental Health Risk Prediction is well-justified. By combining behavioral, psychological, demographic, and environmental factors, the project provides a data-driven and socially impactful solution. It ensures that risks are detected early, interventions are well-targeted, and the overall mental health of individuals is improved.